NANDINI V

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EDUCATION

Innomatics Research Labs 08/2025 – Present
Data Science Hyderabad, India

Karunya Institute of Technology and Sciences 08/2023 – 05/2025
M.Sc. Artificial Intelligence and Data Science Coimbatore, India
CGPA: 7.9

Govindammal Aditanar College for Women 06/2020 – 05/2023
B.Sc. Computer Science Tiruchendur, India
CGPA: 8.6

SKILLS

Programming
Python

Deep Learning
ANN, CNN, RNN, Transformers, Attention Mechanism

Gen AI & LLM
LLM Evaluation Pipelines, RAG, LLM , Fine Tuning, Prompt Engineering, NLP

Frameworks & Libraries
Tensorflow, Langchain, Hugging Face, Scikit-learn, Pandas, Numpy, Matplotlib, sentence-transformers, Ollama, ChromaDB, Gradio, Streamlit

Machine Learning
Supervised Learning, Classification, Regression, Clustering, Model Evaluation, Hyperparameter Tuning, PCA

Tools & Platforms
GitHub, Google Colab, VS Code

PROJECTS

LLM-Powered Legal Case Analyzer
Python, LangChain, ChromaDB, HuggingFace, Ollama, Streamlit
- Designed a production-scale RAG evaluation pipeline for legal document summarization and context-aware Q&A, benchmarking retrieval accuracy across chunk size and embedding strategy variations to establish reproducible evaluation rubrics.
- Built an end-to-end NLP data pipeline — PDF parsing, text chunking, embedding generation, and vector indexing — ensuring consistent, reliable LLM responses without external API dependencies via Ollama-based local inference.
- Detected and resolved retrieval inaccuracies and memory overhead during embedding generation through intelligent chunking and overlap strategies — improving model output consistency and reducing latency at scale.

AI-Based Travel Planning Chatbot
Python, LangChain, NLP, Conversational AI, Prompt Engineering
- Built a conversational AI system for personalized travel planning, designing NLP workflows to handle diverse user queries with consistent, context-aware responses — applying prompt engineering to standardize itinerary generation outputs.
- Implemented conversational memory and context retention mechanisms to maintain dialogue consistency across multi-turn interactions, directly addressing LLM output consistency challenges in generative AI systems.
- Evaluated and improved recommendation quality by auditing model outputs for relevance and personalization accuracy, reducing inconsistent recommendations through iterative rubric-based refinement.

Flight Fare Price Prediction using ANN
Python, TensorFlow, Keras, Pandas, NumPy, Scikit-learn
- Designed and benchmarked an ANN regression model for flight ticket price prediction on 400,000+ rows of Indian aviation data, running systematic experiments across learning rates, epochs, and hidden layer configurations to establish optimal model performance baselines.
- Diagnosed and resolved model drift caused by noisy airline data — applied normalization, outlier handling, dropout layers, and regularization techniques, reducing overfitting and improving prediction generalization across unseen data.
- Performed hyperparameter tuning and feature selection experiments, documenting evaluation methodology and results to ensure reproducible, defensible model performance metrics.

CERTIFICATIONS

Google Advanced Data Analytics
Coursera

Google Al Essentials
Google Cloud

Python Programming
Innomatics Research Labs